FORMULAE LIST

Circle

$x^2 + y^2 + 2gx + 2fy + c = 0$ represents a circle centre $(-g, -f)$, radius $\sqrt{g^2 + f^2 - c}$.

$(x - a)^2 + (y - b)^2 = r^2$ represents a circle centre (a, b) and radius r.

Scalar product

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos\theta, \text{ where } \theta \text{ is the angle between } \mathbf{a} \text{ and } \mathbf{b}$$

or

$$\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3 \text{ where } \mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \text{ and } \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

Trigonometric formulae

$$\begin{aligned}
\sin(A \pm B) &= \sin A \cos B \pm \cos A \sin B \\
\cos(A \pm B) &= \cos A \cos B \mp \sin A \sin B \\
\sin 2A &= 2 \sin A \cos A \\
\cos 2A &= \cos^2 A - \sin^2 A \\
&= 2\cos^2 A - 1 \\
&= 1 - 2\sin^2 A
\end{aligned}$$

Table of standard derivatives

$f(x)$	$f'(x)$
$\sin ax$	$a \cos ax$
$\cos ax$	$-a \sin ax$

Table of standard integrals

$f(x)$	$\int f(x)\, \mathrm{d}x$
$\sin ax$	$-\frac{1}{a} \cos ax + C$
$\cos ax$	$\frac{1}{a} \sin ax + C$

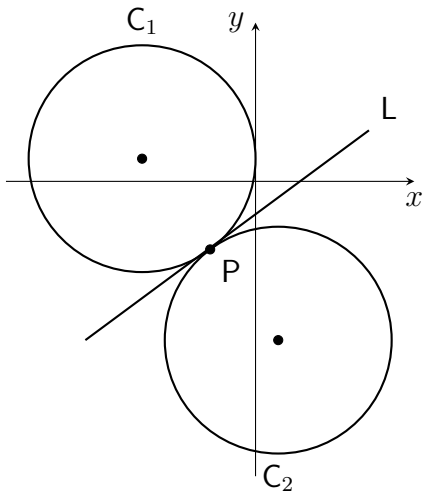
1. Find $\int (5x-7)^{-3}\,\mathrm{d}x$ (2)

2. Simplify $\frac{1}{2}\log_a 9 + \log_a 4$ where $a > 0$. (2)

3. (a) Express $2\sin x^\circ - 5\cos x^\circ$ in the form $k\sin(x-a)^\circ$, $k > 0$, $0 < a < 360$. (4)

   (b) Hence solve the equation $2\sin x^\circ - 5\cos x^\circ = 3$ where $0 < x < 360$. (3)

4. Circles C_1 and C_2 touch externally at point P, as shown in the diagram below.



Circle C_1 has equation $(x+5)^2 + (y-1)^2 = 25$.

(a) Find the centre and radius of circle C_1. (2)

Circle C_2 has equation $x^2 + y^2 - 2x + 14y + 25 = 0$.

(b) Find the centre and radius of circle C_2. (2)

Line L is a common tangent to the two circles at point P.

(c) Determine the equation of line L. (4)

5. Solve $\log_4 32 - 3\log_4 x = 1$ where $x > 0$. (3)

6. Functions f and g are defined on suitable domains by:

   - $f(x) = \sin x$
   - $g(x) = 3x - \frac{\pi}{4}$

   (a) Find an expression for $h(x)$ where $h(x) = f\left(g(x)\right)$. (2)

   (b) Find $h'\left(\frac{\pi}{3}\right)$. (2)

7. Given that: (4)

   - $\dfrac{\mathrm{d}y}{\mathrm{d}x} = (2x-3)^3$
   - $y = 1$ when $x = 2$.

   Express y in terms of x.

8. The level confidence, C %, a Higher Maths student may be expected to feel about the prospect of solving an Exponential Growth question after t hours spent revising and practising the skill may be modelled using the equation:

$$C = 20e^{0.563t}$$

   (a) State the student's initial level of confidence with Exponential Growth. (1)

   (b) Find the time it is expected to take for the student's confidence to double. (3)

END OF QUESTIONS

ANSWERS - Practice Exam F

1. $-\dfrac{(5x-7)^{-2}}{10} + \mathrm{C}$
2. $\log_a 12$
3. (a) $\sqrt{29}\sin(x-68.2)^\circ$
   (b) $x = 102.1, x = 214.3$
4. (a) radius $= 5$ and centre $(-5, 1)$
   (b) radius $= 5$ and centre $(1, -7)$
   (c) $y = \frac{3}{4}x - \frac{3}{2}$ or $4y = 3x - 6$
5. $x = 2$
6. $-\frac{\sqrt{3}}{\sqrt{2}}$ or equivalent
7. $y = \frac{(2x-3)^4}{8} + \frac{7}{8}$
8. (a) 20%
   (b) 1.23 hours